SATYA BONDA

Principal AI Engineer | Principal Agentic AI Architect | Forward Deployed Engineer — Multi-Agent Systems · Enterprise AI Platforms · MCP · RAG · LLMOps · Evaluation-Gated Delivery · Responsible AI · Azure & AWS
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SUMMARY

Principal AI Engineer and forward-deployed agentic AI architect taking enterprise workflows from first customer call to governed production: multi-agent systems on MCP, deterministic code where money is involved, and evaluation harnesses that gate every release and re-test production nightly. Claude Certified Architect – Professional (CCAR-P), Microsoft TITAN Frontier Transformation Engineer, 25+ agentic systems, 18 years leading enterprise delivery.

EXPERIENCE

Principal AI Engineer

Independent AI Consultant · Bengaluru, India · Mar 2026 – Present

Enterprise AI Playground — Solo-Built Governed Agentic AI Platform (LangGraph · MCP) · Live · Repo · Portfolio

- **What it does:** one governed place to try 30 models (11 providers), build and run agents, and see every dollar.
- **How it works:** OpenAI-compatible FastAPI gateway with LiteLLM cost-aware routing; identity, role, budget and key checks before every call and a cost-ledger row after it; LangGraph runs that pause for human review; 175 agent blueprints; a 7,500-server MCP marketplace.
- **How it is proven:** a five-level evaluation ladder (Draft → Golden → Gated → Canaried → Production) with LLM-judge rubrics, nightly canaries and drift rollback.

Flick — Autonomous Invoice Reconciliation (LangGraph · MCP · Claude · Azure) · Apr – May 2026

- **What it does:** replaces 30–90 minutes of analyst work per supplier invoice (proving each line against contracts, POs and meter data) with a Teams chat; in pilot, 1,200 invoices a month at 87% first-pass, 4.2 min vs 47 manual.
- **How it works:** a LangGraph manager routes typed state through Parser (40+ formats, F1 0.94), Matcher, Anomaly Detector, Reconciler and Reporter over five MCP servers; a Critic on a second LLM vendor rejects uncited claims; green auto-posts, amber goes to analysts, red drafts a dispute a human releases.
- **How it is proven:** a 200-invoice analyst-labeled golden set gates every change overnight; 5% canary with auto-rollback; idempotent ledger writes, circuit breakers, model fallback, PII redaction.

Compass — Live Dynamics 365 Agent (Azure AI Foundry · MCP · Copilot Studio) · Jun 2026 – Present

- **What it does:** staff ask about orders, stock, balances or store procedures in plain English and get a cited, company-scoped answer from live Dynamics 365 F&O in Teams, web, MCP tool packs and D365 Agent Factory.
- **How it works:** a schema-constrained GPT-4.1 planner over 39 read-only OData entities plus Azure AI Search over the business's own SOPs; every figure computed in code; access never exceeds the user's F&O role.
- **How it is proven:** an 88-case battery and an LLM judge (95% human agreement) test production nightly; functional packs green every night since Sep; red nights auto-file GitHub issues; 12 OWASP adversarial cases.

Quote Intelligence — Fabrication-Proof Quote Comparison (Agents SDK · MCP · React) · Apr 2026

- **What it does:** turns a home builder's quote and a competitor's into a source-traced, cent-exact comparison (web, Excel, Word) in 6–8 minutes instead of half a day, with no invented prices.
- **How it works:** page-anchored extractors, alignment, ambiguity flagging and a different-model verifier feed a pure-code tripwire checking each snippet against its cited page; Decimal math; unconfirmed rows block the total.
- **How it is proven:** 63 assertions in four suites (26 adversarial) on every change, zero fabricated figures; verified end-to-end on a live customer job (57 rows, 26 ambiguities flagged); IDOR/SSRF guards, no customer PII.
- **Agentic RAG on Bedrock:** self-correcting LangGraph agent, hybrid retrieval, eval-gated CI/CD, MCP server.

AI Program Manager / Agentic AI Architect

Insight Global · Bengaluru, India · May 2024 – Feb 2026

- **Fortune-100 AI transformation:** ran four AI programs for a global technology company's Credit & Collections organization — payment insights, a governed Finance Bot Store in Teams and an agentic AI co-worker — through steering committees and executive updates; 2× accounts under AI, 71% utilization, $220M collected.
- **SOX 404 agentic co-worker (prototype to MVP):** a validation plan authored once per control, run identically every period; no claim ships without cited evidence; golden-set evals and SME gates; proven on three controls.

- **Public portfolio:** 25+ eval-gated agentic systems across finance, healthcare, voice and supply chain.
- **Enablement:** Claude Certified Architect prep and agentic AI cohorts at EY; GenAI, ML and analytics training for Bristlecone, National Grid and Navikenz.

EARLIER EXPERIENCE

Wipro | Project Manager / AI Engineer · Jul 2021 – Apr 2024

- **Product ownership:** owned the Oracle CPQ/MDM roadmap; modernization saved ~$900K a year and 3 FTE.
- **GenAI and ML engineering:** conversational RAG assistant (LangChain, FAISS, GPT-4) and LLM fine-tuning (MarianMT, BLOOM LoRA) evaluated on BLEU and COMET; Wipro AI Hackathon 2024 winner.

Trane Technologies / Ingersoll Rand | Project Manager & Agile Coach · Jun 2015 – Jul 2021

Oracle 19c upgrade under budget; zero-disruption ERP separation (President Award); promo-claims CSAT +40%.

Volvo | IT Project Manager · 2013 – 2015 · Oracle | Project Management Consultant · 2011 – 2013

IBM | Project / Operations / Transition / Quality · Jan 2007 – Apr 2011

Ran 5 helpdesks (55 FTE) and service-desk transitions across 4 clients, with audits and compliance.

Genpact | Process Leader · Dec 2005 – Jan 2007 · HP | Quality Specialist · Jan 2004 – Nov 2005

SKILLS

Agentic AI & Multi-Agent Orchestration: Agentic AI, multi-agent systems, supervisor–worker, handoff meshes, generator–critic, fan-out/fan-in, ReAct, tool calling, human-in-the-loop, checkpointing, model routing, verifier gates.

Claude Engineering: Claude API, Claude Agent SDK, Claude Code, context engineering, prompt engineering, prompt caching, prompt versioning.

Frameworks & Protocols: Model Context Protocol (MCP), FastMCP, A2A, LangGraph, LangChain, OpenAI Agents SDK, CrewAI, AutoGen, PydanticAI, Microsoft Agent Framework, Google ADK, LiteLLM, Langflow, n8n, FastAPI, Next.js (React), TypeScript, Node.js, Streamlit.

GenAI & Retrieval: RAG, GraphRAG, agentic and corrective RAG, hybrid search (BM25 + vector), reranking, embeddings, chunking, vector databases (Pinecone, pgvector, FAISS, ChromaDB, Azure AI Search, S3 Vectors); fine-tuning (LoRA, QLoRA, SFT, DPO).

Evaluation, Observability & Reliability: Evaluation harnesses as release gates, golden datasets, LLM-as-a-Judge, adversarial suites and red teaming, RAGAS, DeepEval, LangSmith, OpenTelemetry, AI observability, monitoring, drift detection, canary + auto-rollback, circuit breakers.

Machine Learning: Python, SQL, PostgreSQL, PyTorch, TensorFlow, scikit-learn, XGBoost, Transformers.

Cloud & AI Platform Engineering: Azure OpenAI, Azure AI Foundry, Foundry Agent Service, App Service, Container Apps, Key Vault, Entra ID, Copilot Studio, Power Platform, Dynamics 365 F&O, Databricks; AWS: Amazon Bedrock, Bedrock AgentCore, Lambda, ECS Fargate, S3, DynamoDB.

LLMOps / MLOps & CI/CD: MLflow, model registry, CI/CD for models, Kubernetes, Docker, scalable serving (Container Apps, auto-scaling ECS Fargate), Terraform/Bicep, AWS CDK, GitHub Actions, blue/green deploys, Pytest, TDD, code reviews, architecture decision records, DORA metrics.

AI Security, Responsible AI & Governance: Responsible AI, EU AI Act, PII/PHI redaction, Content Safety, prompt-injection defense, OWASP LLM Top 10, Managed Identity, SSO (Okta, OIDC), role-based access, audit trails.

AI Product & Leadership: AI strategy and roadmaps, AI product management, turning ambiguity into plans, stakeholder and executive communication, AI adoption and ROI, AI FinOps, team leadership, mentoring.

CERTIFICATIONS

Anthropic Claude Certified Architect – Professional (CCAR-P), 2026 · Microsoft Agentic AI Business Solutions Architect – Expert (AB-100, 2026) · Microsoft Azure AI Apps & Agents Developer Associate (AI-103, 2026).

Microsoft TITAN Badge – Frontier Transformation Engineer (2026).

Also: PMP, PMI-RMP, CSM, SAFe POPM, ITIL V3, Azure Solutions Architect Expert, AWS Cloud Practitioner, Oracle GenAI Professional, Oracle ML Associate.

EDUCATION

Master of Computer Applications (MCA) — Periyar University, 2003

B.Sc. Physics (Hons.) — University of Calcutta, 1999

PROJECT PORTFOLIO — SOURCE CODE AND LIVE DEMOS

Bedrock Agentic RAG — Self-Correcting RAG on AWS (LangGraph · Amazon Bedrock · MCP)

LangGraph agent on Amazon Bedrock that grades its own retrieval, rewrites missed queries and checks groundedness; S3 Vectors + BM25 with RRF and Bedrock Rerank; Lambda ingestion, auto-scaling ECS Fargate, DynamoDB checkpoints, AWS CDK; eval-gated CI/CD (hit@5 1.00, MRR 0.94); MCP server. Repo

IQR — Agentic Audit-Evidence Validation (Azure AI Foundry)

Multi-agent SOX-control validation with a blinded verifier and citation-gated verdicts; five-gate eval harness (defect recall, false exceptions, citation validity, abstention, reproducibility) at 100% stability; per-seat model routing. Repo

Dentsi — Autonomous AI Front Desk (Multi-Agent Voice AI)

Six-agent voice front desk: GPT-4 orchestrator, 18-tool loop for caller ID, triage and scheduling (Twilio, ElevenLabs, Deepgram, NestJS); HITL escalation; PHI consent logging. Repo · Live

AI Sales Enablement Platform (MCP Server Federation)

CRM, analytics, RAG and training capabilities as four MCP servers (~20 tools) behind a FastAPI gateway; Streamlit console and Next.js dashboard; transcript pipeline feeding the RAG layer. Repo · Live

Autonomous Email Agent (LangGraph)

Hierarchical StateGraph: triage router to a Gmail tool-calling subgraph (GPT-4.1) on LangGraph Platform with Agent Inbox; interrupt HITL at triage and tool level, Store-backed memory learning from human edits, cron ingestion, golden-dataset evals with GPT-4o judge. Repo

Agentic Invoice Processing (Supervisor–Worker)

Nine-agent supervisor-worker: extraction, then parallel validation fan-out to contract, MSA and master-data agents; quality-review and learning agents close the loop; HITL escalation. Repo · Live

Autonomous Document Anomaly Detection (Multi-Agent Pipeline)

Ingests leasing contracts, invoices and POs; flags anomalies with per-finding confidence; dual-orchestrator, parallel processing, confidence-thresholded review gate. Repo · Live

Supply Chain Intelligence Platform (Multi-Agent Handoff Orchestration)

Six-agent Agents SDK platform: orchestrator handoffs to risk, demand, supplier, inventory, logistics and reporting specialists over 15 deterministic calculators; model-free computation; e2e suites. Repo · Live

PMO CoPilot (Twelve-Agent Handoff Mesh)

Twelve-agent Agents SDK portfolio assistant: orchestrator handoffs to eleven specialists (status, EVM, risk, escalation, scheduling, resources, milestones, analytics, reporting); deterministic CPI/SPI; ML risk predictions. Repo · Live

Domain-Specific LLM Fine-Tuning — EN to NL Translation (LoRA + Full Fine-Tune)

Full-parameter MarianMT fine-tuning (Lightning) and BLOOM-560m LoRA (1.2M trainable, 0.2%); evaluated on FLORES + domain set with BLEU, chrF, TER, COMET. Repo · Live

Additional Shipped Systems

- **SDLC Agents Platform —** four-agent CrewAI pipeline + FastAPI + ChromaDB; live console. Repo · Live
- **Multi-Model Research Agent (MCP) —** Tavily-to-Serper failover; tri-provider fallback; MCP tools. Repo · Live
- **Agentic Workflow Automation —** role-based Agents SDK agents to ClickUp behind HITL gates; FastMCP server. Repo · Live
- **Agentic CoLearn —** curriculum-grounded generator-critic; BKT/IRT/FSRS mastery. Repo · Live
- **DentaScribe —** five-stage clinical scribe; transcript-grounded SOAP + CDT; 102 tests. Repo · Live
- **Sage Mind —** AutoGen tutor; nested five-reviewer critic panel gates every draft. Repo · Live
- **Healthcare Prior Authorization —** five-agent pipeline: CPT/ICD-10 extraction; denial-to-appeal branch. Repo · Live
- **Copilot Studio Agent Client —** Entra ID (MSAL) SPA; conversations API, tool-trace panel, Adaptive Cards. Repo